EXERCISE 04

# Step Turning and Chamfering

## Aim

- To perform step turning and chamfering operations to produce a cylindrical part of given dimensions.

### Required instruments

1. Facing tool
2. Plain turning tool
3. Chamfering tool
4. Chuck key
5. Tool holder
6. Steel rule
7. Vernier caliper
8. Outside caliper
9. Inside caliper
10. Surface gauge

## Tool setting

A facing tool, a plain turning tool and a chamfering tool are mounted on the toolpost and their cutting edges are set exactly aligned to the axis of the lathe spindle. The compound rest is set at $0^0$ mark.

## Holding of the work

1. The metal piece to be machined is held in the chuck of the lathe.
2. Surface gauge is used to check whether the work is held aligning with the lathe axis.

## Selection of cutting speed

$$\text{Cutting speed} = \frac{\pi DN}{1000} \text{ m/min.}$$

The above formula is used to arrive at a proper cutting speed for machining and the spindle speed is adjusted suitably.



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